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Lab 04.2: DRISMI Experience

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Abstract

A test session was conducted to physically perceive the two **mode shapes** identified in the quarter car model:

- **Mode 1 (Low-Frequency):** characterized by large-amplitude movement of the entire cabin/chassis, visible via hydraulic actuators
- **Mode 2 (High-Frequency):** perceived as small-amplitude vibrations through the seat and floor pan, such as road texture
- **Fidelity:** while distinct mechanically, these modes are difficult for a driver to differentiate during transient events like road bumps

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1. Introduction

This report summarizes observations made during a test session within a vehicle simulator, with the aim to feel the first two mode shapes of a road vehicle, highlighted in the quarter car model. Analysis is conducted focusing on the perceived differences and the external visual confirmation of system performance.

2. Quarter car model modes

1. **Mode 1 (Low-Frequency/Large displacement of sprung mass):** In this mode, the entire chassis or cabin of the simulator was observed to move. This mode is typically used to generate large-amplitude, sustained vibrations. Externally, this motion was clearly visible via the hydraulic actuators located beneath the cabin platform.

2. **Mode 2 (High-Frequency/Large displacement of unsprung mass):** This mode was characterized by a vibration felt from beneath the seat and floor pan. This effect is primarily intended to transmit high-frequency, small-amplitude vibrations (e.g., road texture, engine vibration, or impacts). External observation confirmed this nature, as a high-frequency vibration of the driver’s head was noted via the control room webcam.

3. Conclusion

The ability to clearly distinguish between Mode 1 and Mode 2 when traversing road bumps was complicated. This difficulty suggests that high-amplitude, transient events may mask or blend the low-frequency chassis movement and the high-frequency haptic feedback. However, observing the simulator from the control room provided clear distinction between the modes: the large-scale displacement of the actuators corresponded to Mode 1, while the rapid, subtle head vibration captured on the webcam confirmed the activation and effect of Mode 2.